Multimodal RAG Platform - Architecture Design Document

Version 2.4 - Platform Team

1. Executive Summary

This document specifies the caching architecture agreed during the quarterly architecture review led by Sarah Chen (Principal Engineer). It accompanies the recorded meeting session and the annotated caching diagram distributed to the team.

2. Background and Problem Statement

PostgreSQL read traffic grew 300% over the last quarter.
Peak-hour CPU utilization on the primary replica reached 85% and P95 query latency exceeded 1.2 seconds, violating our 200ms target.

Root cause: repeated execution of hot read queries that change rarely, including product catalog lookups and user session joins.

3. Proposed Solution: Redis Caching Layer

Section 3.2 - Data Flow
A Redis cache sits between application servers and PostgreSQL.

Read path: App Server -> Redis -> (on miss) -> PostgreSQL,
then populate Redis with TTL of 300 seconds (5 minutes).
Write path: App Server -> PostgreSQL first, then synchronously
invalidate and refresh the matching Redis key
(write-through pattern).

Expected impact: about 60% of read queries served from Redis, reducing database load proportionally. See Figure 7-1, appendix page 7 for the complete annotated data-flow diagram discussed in the meeting.

4. Operational Notes

QPS target: 10,000 sustained. SLA: 99.9% availability.
If Redis is unavailable all traffic falls back to PostgreSQL;
Redis sits on the performance critical path, not the availability path.

Appendix reference: Figure 7-1 on page 7 mirrors the slide shown at minute 00:40 in the recorded review session.

7. Appendix: Data Flow Diagram (Figure 7-1)

Redis + PostgreSQL Caching Architecture

```
[Clients] -> [API Gateway] -> [Load Balancers] -> [App Servers]
                                    /          \
                      [Redis Cache] <           > [PostgreSQL]
                        TTL 300s              write-through
```

Annotations:

- Read reduction target: 60%
- TTL: 5 minutes
- Proposed by: Sarah Chen, Principal Engineer
- Presented in meeting_recording.mp4 at 00:40